Tortho–SatGS: A 3-D Gaussian Splatting-Based Method for True Orthophoto Generation From Multiview Satellite Imagery

Junxing Yang, Wenyao Bian, Xingchen Liu, Shuo Gao, Yingjie Qu, and He Huang

***Abstract*—True digital orthophoto maps (TDOMs) accurately represent the true spatial positions and visual appearances of ground objects at the urban scale, and constitute a fundamental remote sensing product for large-scale urban digitization and fine-grained geographic modeling. Conventional TDOM generation methods that rely on digital elevation models (DEMs) or digital surface models (DSMs) are highly susceptible to occlusions and elevation errors in complex urban environments, often resulting in geometric distortions and visual artifacts. Neural radiance field (NeRF)-based approaches can mitigate these issues, but their high computational cost limits practical deployment on large-scale satellite imagery. To address these challenges, we present Tortho–SatGS, a pure vision-based framework that, to the best of our knowledge, is among the first to systematically integrate 3-D Gaussian Splatting (3DGS) into satellite true orthophoto generation. Specifically, we design a 3DGS-based geometric modeling pipeline tailored to satellite imaging geometry and introduce an orthographic rasterization-based rendering scheme to effectively resolve building side facades and edge curvature artifacts, enabling DEM/DSM-free TDOM generation with improved geometric consistency. In addition, a VGG-based perceptual loss is incorporated to complement pixel-level supervision, improving radiometric consistency and fine-grained texture fidelity, particularly in shadowed and low-texture regions. Experimental results on two real-world satellite datasets demonstrate that Tortho–SatGS consistently outperforms conventional methods in terms of geometric accuracy, texture quality, and radiometric consistency. Compared with NeRF-based approaches, our method achieves approximately 27× faster training speed and 15× faster rendering speed, effectively balancing reconstruction accuracy and computational efficiency. These results validate the effectiveness and practical advantages of 3DGS-based approaches for the generation of true orthophotos of satellites.**

***Index Terms*—3-D Gaussian Splatting (3DGS), satellite imagery, true digital orthophoto map (TDOM), urban remote sensing.**

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I. Introduction

ACCURATE orthophoto imagery provides a unified and reliable spatial reference for a wide range of geospatial applications, including urban planning, resource investigation, and environmental monitoring. The quality of orthophotos directly affects the accuracy of subsequent measurement, modeling, and spatial analysis. Therefore, the efficient generation of orthophoto imagery with both high geometric precision and visual consistency has long been a fundamental problem in remote sensing image processing and geographic information modeling. In recent years, remote sensing observation and intelligent interpretation technologies have advanced rapidly. Hyperspectral remote sensing enables the joint acquisition of spatial and spectral information, providing richer discriminative features for fine-grained scene representation and quality enhancement of remote sensing products [1]. Meanwhile, recent progress in large-scale remote sensing data processing and analysis has significantly improved the exploitation of multisource satellite imagery, establishing a stronger technical foundation for high-precision orthophoto generation from multiview observations [2]. Compared with aerial and uncrewed aerial vehicle (UAV) imagery, satellite imagery offers significant advantages in terms of coverage, revisit frequency, and diversity of imaging conditions, making it an important data source for large-scale, high-quality orthophoto generation. However, satellite imaging is usually associated with more complex off-nadir viewing geometry, more pronounced cross-view radiometric variations, and much larger scene scales, which make high-quality true digital orthophoto map (TDOM) generation from multiview satellite imagery still a challenging task.

Specifically, multiview satellite TDOM generation is jointly constrained by four key aspects: geometric modeling, occlusion handling, radiometric consistency, and large-scale scalability. First, variations in viewing geometry, sensor model inaccuracies, and uncertainties in orbital and attitude parameters can introduce spatial distortions and local misalignments during reprojection and registration. Second, high-rise buildings in complex urban scenes create severe 3-D occlusion relationships, and inaccurate elevation estimation or unstable visibility reasoning often leads to geometric artifacts such as facade residues and edge bending. Third, shadow distribution, solar elevation changes, and surface reflectance differences result in brightness and color inconsistencies across views, making it difficult to recover natural textures in shadowed

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regions, water bodies, and other weak-texture areas using only pixel-wise consistency constraints. Finally, large-scale, high-resolution satellite scenes impose substantial demands on modeling efficiency, inference speed, and memory consumption, and balancing geometric accuracy, visual quality, and reconstruction efficiency remains a key challenge for existing methods.

Existing TDOM generation methods can generally be grouped into three categories: geometry-based transformation methods [3], [4], [5], [6], [7], interpolation-based approaches [8], [9], [10], and deep learning-based image reconstruction methods [11], [12], [13], [14], [15], [16], [17]. Traditional methods typically rely on external priors such as digital elevation models (DEMs) and digital surface models (DSMs) for image reprojection, together with occlusion detection and texture compensation to generate true orthophotos. Although such methods can achieve reasonable geometric performance when accurate 3-D priors are available, their effectiveness strongly depends on the quality of external elevation data, whose acquisition and preprocessing are often costly. In densely built or topographically complex urban environments, elevation errors are difficult to avoid and may further degrade the stability of occlusion detection and texture compensation. Texture completion methods [18] attempt to fill occluded regions by introducing visible textures from neighboring views, but due to the sparse viewing angles and discontinuous texture information inherent to satellite imagery, they still suffer from limited completion accuracy and poor texture consistency.

In recent years, advances in novel view synthesis (NVS) techniques [19] have provided a new perspective for TDOM generation. As a representative NVS method, neural radiance fields (NeRF) [20] model scene geometry and appearance through a continuous implicit radiance field, enabling the synthesis of high-fidelity novel views from multiview imagery. NeRF has shown strong potential in preserving building boundaries and improving texture consistency. To accommodate the large scene scales, complex imaging geometry, and illumination variations of satellite imagery, several satellite-adapted extensions of NeRF have been proposed [21], [22], [23], [24]. Previous studies [11], [25], [26], [27], [28], [29] have demonstrated that NeRF-based TDOM generation can effectively alleviate building distortions and occlusion-related artifacts that are difficult to avoid in traditional methods. However, these methods usually require dense sampling and volumetric rendering along each ray, resulting in high computational cost and limited scalability for large-scale satellite scenes. Moreover, the implicit continuous representation lacks more direct explicit geometric constraints, which still limits geometric consistency modeling, structural representation, and subsequent scene editing.

In summary, although existing multiview satellite TDOM generation methods have made progress in external elevation-based correction, occlusion completion, and implicit radiance field modeling, they still struggle to simultaneously achieve occlusion robustness, geometric consistency, radiometric stability, and large-scale processing efficiency in complex urban environments. Therefore, how to directly learn a scene representation from multiview satellite imagery that satisfies true orthographic projection, incorporates explicit geometric constraints, and maintains radiometric consistency, without relying on high-precision external elevation priors, remains a key problem in satellite TDOM generation.

To address the above issues, this article proposes Tortho–SatGS, a satellite true orthophoto generation method based on 3-D Gaussian Splatting (3DGS) [30], [31]. Unlike traditional methods that rely on DEM/DSM and explicit occlusion detection, the proposed method directly learns an explicit 3-D Gaussian representation from multiview satellite imagery. Compared with NeRF-based approaches that depend on implicit volumetric rendering, we introduce a satellite-oriented geometric modeling strategy and an orthographic rasterization-based rendering scheme, enabling the rendering process to more directly satisfy orthographic projection constraints and effectively suppress geometric artifacts such as facade residues, edge bending, and local misalignments. Benefiting from the explicit Gaussian representation and efficient rasterization, the proposed method significantly improves both training and rendering efficiency. In addition, a VGG-based perceptual loss is incorporated to further enhance radiometric consistency and fine-grained texture fidelity in shadowed and low-texture regions. The main contributions of this article are summarized as follows.

1) We propose Tortho–SatGS, a novel TDOM generation framework tailored for satellite imagery. To the best of our knowledge, this is the first pure vision-based approach that systematically integrates 3DGS into the satellite TDOM generation task. Through unified geometric–radiometric modeling and rendering, the framework enables high-quality TDOM generation without relying on auxiliary data such as DEM, DSM, or DBM.
2) We optimize the 3DGS framework for satellite imaging characteristics by introducing a geometry modeling strategy and a differentiable projection transformation that efficiently converts perspective projection into orthographic projection, overcoming the inherent limitations of applying 3DGS to satellite remote sensing scenarios.
3) We incorporate a VGG-based perceptual loss to complement pixel-level supervision, improving radiometric consistency and fine-grained texture fidelity in weakly constrained regions such as shadowed and low-illumination areas.

II. Related Works

A. Traditional TDOM Generation

The fundamental distinction between true digital orthophotos and conventional orthophotos lies in the treatment of nonground objects. While traditional orthorectification primarily corrects geometric distortions caused by terrain relief, true digital orthophotos further eliminate occlusions and tilt-induced projection effects introduced by elevated objects such as buildings, ensuring that each image pixel strictly corresponds to its true ground position. Consequently, TDOM generation essentially involves accurate modeling and resolution of imaging geometry and spatial occlusion relationships,

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typically relying on DSM, digital terrain models (DTMs), and the 3-D projection geometry defined by exterior orientation parameters. As a core data source, high-quality DSMs are usually obtained through multiview satellite 3-D reconstruction or stereo matching pipelines, for which a relatively systematic research framework has been established [32], [33], [34]. Depending on the adopted stereo matching strategy, traditional multiview satellite reconstruction methods can be broadly categorized into multiview matching (MVM), multistereo matching (MSM), and deep learning-based stereo matching (DLSM) approaches [32]. Specifically, MVM directly generates dense disparity maps through global MVM to recover scene geometry, MSM decomposes multiview reconstruction into multiple stereo pairs followed by fusion, and DLSM leverages deep neural networks to improve matching accuracy in complex scenes [35]. These reconstruction techniques provide high-quality DSMs that serve as a fundamental basis for traditional TDOM generation. For occlusion detection, the Z-buffer method [36] is among the earliest visibility analysis algorithms applied to DSM-based TDOM generation. This approach constructs a depth buffer and a visibility matrix to record the distance between surface elements and the camera center, as well as the visibility status of DSM grid cells. Occlusions are determined through depth comparisons along the viewing direction. Due to its computational simplicity and implementation efficiency, the Z-buffer method has been widely adopted in photogrammetry and computer vision. However, it is highly sensitive to DSM resolution and often suffers from false occlusions, false visibility, and multioverlapping fragments, which degrade occlusion detection accuracy. To address these limitations, various improved methods [37], [38], [39], [40], [41] have been proposed. DSM sorting-based occlusion detection methods [40] introduce elevation-priority ordering into the Z-buffer framework, using height information to infer occlusion relationships, thereby partially improving memory usage and computational efficiency. In contrast, angle-based occlusion detection methods [39] determine occlusions by analyzing variations in viewing angles, effectively reducing false occlusions and false visibility. Vector-based occlusion detection approaches [38] further vectorize pixel regions into square elements and incorporate central projection geometry corrections to mitigate misclassification caused by terrain undulations and camera tilting, albeit at the cost of significantly increased computational complexity.

DBM, which provide detailed structural representations of buildings, constitute another important data source for TDOM generation. DBM-based occlusion detection methods [36], [41], [42], [43], [44] project building models into the imaging geometry and analyze the occlusion relationships between buildings and the ground along viewing rays, thereby determining surface visibility from satellite perspectives. Iterative height-surface projection algorithms [42] jointly consider building-to-ground occlusion and interbuilding occlusion but suffer from low computational efficiency, particularly in complex terrain. Amhar et al. [36] employed the Z-buffer method to detect occluded regions and generate mask images, using DTM and masks for terrain orthophoto production and DBM for building orthophoto generation, which were subsequently fused into true orthophotos. The polygon-based imaging (PBI) method [44] enhances robustness in complex urban scenes by combining DRG rasterization with integrated filtering models. The minimum boundary sector (MBS) method [41] replaces traditional minimum bounding rectangles with MBSs, significantly reducing candidate detection regions; however, fundamental issues such as false occlusion and false visibility remain difficult to fully eliminate.

Overall, traditional TDOM generation methods explicitly model imaging geometry and spatial occlusion relationships and can achieve visibility analysis and occlusion correction to a certain extent. Nevertheless, their strong dependence on high-precision 3-D data, along with their sensitivity to occlusion modeling in complex urban environments, continues to impose limitations in terms of efficiency, robustness, and result consistency.

B. NeRF-Based TDOM Generation

NeRF [19] represent a neural network-based approach for 3-D scene reconstruction. By parameterizing a scene as a continuous 5-D function using a multilayer perceptron (MLP), NeRF maps 3-D spatial coordinates and 2-D viewing directions to RGB color and volumetric density. Combined with pinhole camera models and volumetric rendering, NeRF enables high-fidelity reconstruction of detailed 3-D scenes from multiview images, offering an effective and practical solution for traditional 3-D reconstruction and motivating its application to TDOM generation. In the context of UAV imagery, Ortho-NeRF [25] improved sampling strategies and volume rendering logic to address artifacts, holes, nonorthographic projections, and jagged edges commonly observed in traditional methods and mainstream photogrammetric software, achieving accurate UAV-based TDOM generation. Wei et al. [28] extended NeRF-based methods to large-scale urban scenes by optimizing scene representations to accommodate dense buildings and diverse land-cover types, while improving computational pipelines to enhance large-area processing efficiency. To address the low training efficiency of NeRF, Wang et al. [29] integrated multiresolution hash encoding into NeRF-based frameworks, constructing compact and efficient network architectures for fast orthophoto generation over large UAV scenes. Although these approaches are based on conventional NeRF formulations for 3-D reconstruction, the original NeRF assumes a pinhole camera model, which fundamentally differs from the rational polynomial coefficients (RPC) model commonly used in satellite imagery. To bridge this gap, Zhang et al. [45] proposed a local weak-perspective camera approximation of the satellite RPC model, reducing the complexity of modeling satellite imaging geometry and enabling subsequent NeRF-based satellite applications. Building upon this foundation, researchers have introduced various adaptations to NeRF to address large-scale scene modeling, complex imaging geometry, and in-orbit illumination variations. Representative works include S-NeRF [21], Sat-NeRF [22], EO-NeRF [23], sat-mesh [24], and SAT-NGP [46], which incorporate satellite-specific architectural designs and parameter modeling to improve applicability and reconstruction

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robustness. Further advancing NeRF-based TDOM generation, Qu et al. [11] proposed a novel approach that constructs an implicit NeRF and introduces true orthographic volume rendering by casting vertical rays from the top of the scene bounding box to query colors, effectively extending NeRF from perspective to orthographic projection and eliminating terrain displacement and boundary distortion. Wang et al. [47], based on Sat-NeRF, incorporated solar correction terms and depth supervision into the loss function to better handle illumination variations and transient objects, enabling satellite-based DEM generation.

In summary, NeRF-based TDOM research has evolved systematically from UAV-scale sampling and rendering optimizations to satellite-scale adaptations involving camera model approximation, illumination modeling, and network lightweighting. These studies demonstrate the potential of neural representations for remote sensing-based geometric reconstruction and orthophoto generation. However, these approaches still face several limitations in the context of satellite TDOM generation. First, their reliance on dense sampling and volumetric rendering leads to high computational cost, limiting scalability to large-scale scenes. Second, the implicit continuous representation lacks explicit geometric constraints, which can result in suboptimal structural consistency, especially in complex urban environments. Third, the rendering process is inherently inefficient, making it difficult to achieve a practical balance between reconstruction accuracy and processing efficiency. These limitations restrict the applicability of NeRF-based methods for large-scale, high-resolution satellite orthophoto generation.

C. 3DGS-Based TDOM Generation

3DGS has achieved significant breakthroughs in NVS and 3-D reconstruction [48], [49], [50], [51]. By representing scene geometry and radiance using explicit 3-D Gaussian primitives and leveraging fast rasterization-based rendering, 3DGS significantly improves optimization and rendering efficiency while maintaining high reconstruction quality. Recent studies [52], [53], [54] have demonstrated the superiority of UAV-based 3DGS methods in city-scale reconstruction and TDOM quality enhancement, particularly in reducing aliasing artifacts along building edges and enhancing vegetation details compared with implicit NeRF representations. Ortho-3DGS [52] introduced depth supervision obtained from the Depth Anything network to constrain Gaussian parameters and employed orthographic rasterization for TDOM generation, marking the first effective application of 3DGS to orthophoto generation tasks. Tortho-Gaussians [53] further proposed an Orthogonal Splatting mechanism to replace conventional perspective splatting, thereby bypassing DSM construction and occlusion detection, enabling efficient large-scale TDOM generation with great structural detail. Another recent approach [55] combined on-the-fly structure-from-motion (SfM) for real-time pose estimation and sparse point cloud generation with Gaussian sampling and adaptive optimization, achieving near real-time TDOM generation. In the latest developments, the EOGS method [31] extended the 3DGS framework to satellite imagery by approximating pushbroom imaging geometry with an affine camera model and incorporating a shadow mapping module to simulate physical shadow effects, enabling satellite-based 3-D reconstruction. However, EOGS primarily targets 3-D reconstruction, and its model architecture and rendering pipeline are not explicitly optimized for the strict geometric consistency requirements of true orthophoto generation. Motivated by these limitations, this article proposes Tortho–SatGS, which deeply integrates 3DGS principles with satellite imaging mechanisms to achieve high-precision and high-efficiency satellite TDOM generation.

III. Method

In this section, we first describe the satellite dataset processing pipeline adopted in Tortho–SatGS. Then, we introduce the application of 3DGS to satellite imagery. Finally, we present the rendering strategy for generating TDOM. The overall workflow of the proposed method is illustrated in Fig. 1.

A. Image Preprocessing

This study adopts multiview satellite images equipped with rational polynomial coefficients (RPC) as input data. To satisfy the requirements of geometric consistency and radiometric quality imposed by the subsequent 3DGS framework, the raw satellite imagery is standardized through a preprocessing pipeline [11] following the satellite image processing workflow described in Fig. 2.

First, based on the geographic boundaries of the area of interest (AOI), projection correction and region cropping are performed using the embedded RPC models provided with the satellite imagery. This process ensures that images acquired from different viewing angles precisely cover the AOI within a unified geographic reference frame, thereby mitigating geometric discrepancies introduced by satellite attitude variations and orbital errors. Subsequently, a pan-sharpening strategy proposed in [56] is applied to fuse panchromatic and multispectral images, aiming to enhance spatial texture details while preserving spectral information. To balance computational efficiency and compatibility with the subsequent processing stages, tone mapping and dynamic range compression are further conducted following the method described in [30], converting the original 16-bit imagery into an 8-bit representation. After these preprocessing steps, a multiview satellite image dataset that satisfies both geometric consistency and radiometric quality requirements is obtained and subsequently used as input for the 3-D Gaussian primitive modeling stage.

B. SfM-Based Sparse Gaussian Initialization

The original EOGS framework initializes sparse Gaussian primitives by uniformly sampling points in the normalized device coordinate (NDC) space. While this random initialization is generic and unbiased, it does not exploit the strong geometric constraints inherent in satellite imaging, often resulting in floating primitives, background leakage, and unstable optimization in the early training stages—especially for large-scale orthographic rendering. Several works [22], [31],

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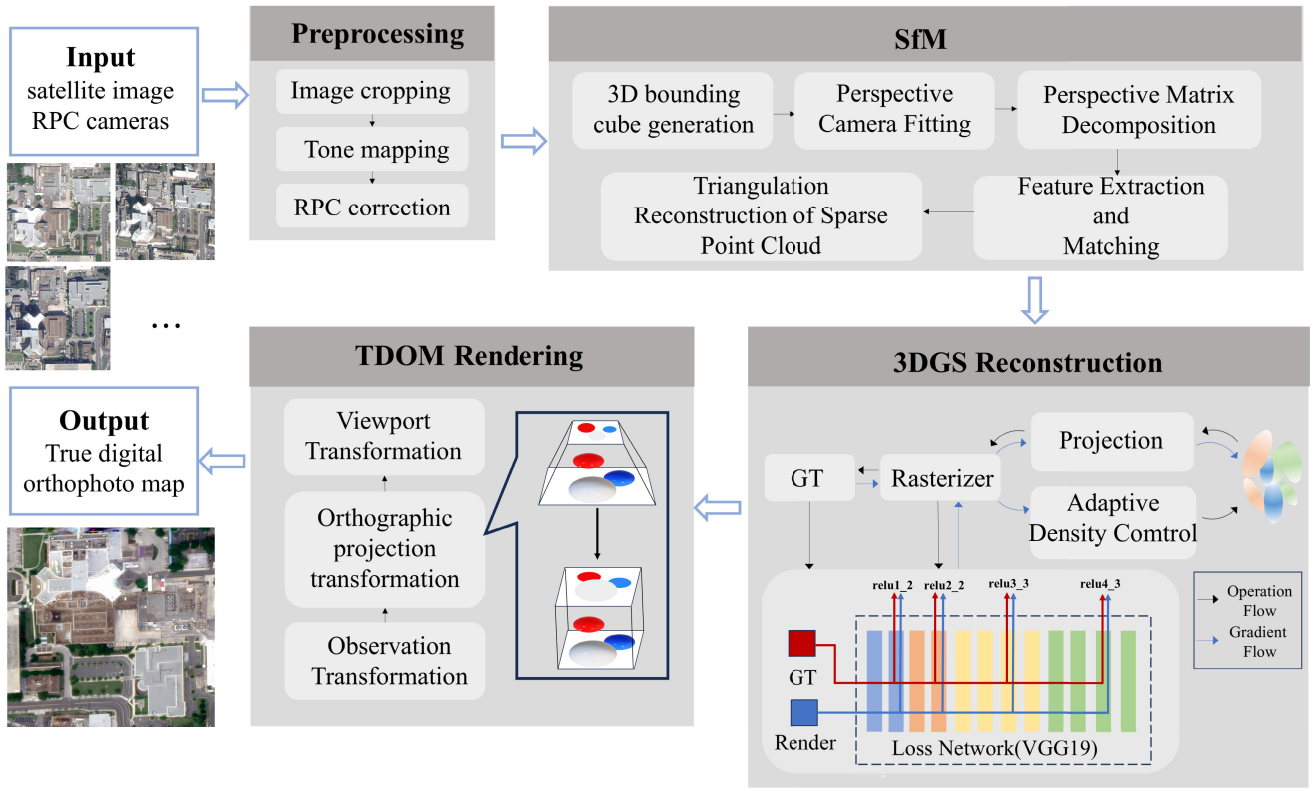


Fig. 1. Overview of the proposed Tortho–SatGS pipeline. High-resolution multiview satellite images are first preprocessed for geometric and radiometric normalization. A satellite-adapted SfM pipeline is then employed to generate sparse geometric priors, which are subsequently converted into 3-D Gaussian primitives for scene reconstruction and orthographic rendering.

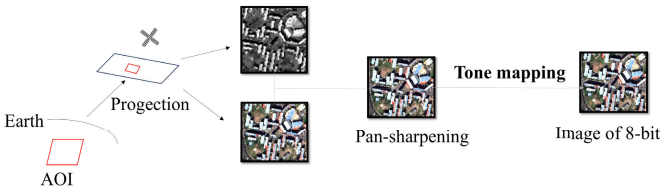


Fig. 2. Workflow of satellite image preprocessing for 3DGS. (From left to right): 1) AOI is projected via RPC-based geometric correction, producing aligned panchromatic (PAN) and multispectral (MS) images; 2) PAN and MS images are fused via pan-sharpening to generate high-resolution color images; and 3) tone mapping converts 16-bit data to 8-bit format, yielding preprocessed multiview images for subsequent 3-D Gaussian modeling.

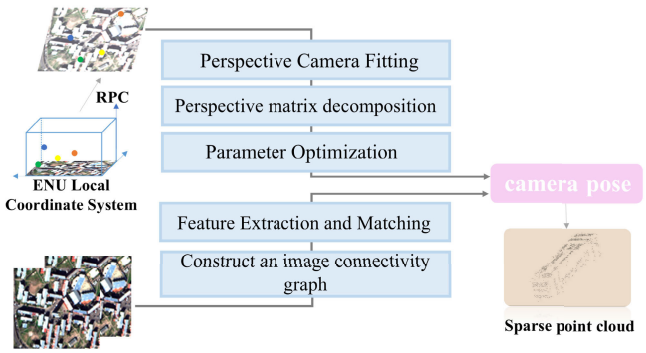


Fig. 3. Satellite-adapted SfM pipeline for sparse Gaussian initialization: 1) perspective camera approximation: RPC model is converted to an equivalent perspective camera via fitting, decomposition, and optimization to obtain accurate camera poses; and 2) sparse point cloud reconstruction: multiview images are processed through feature matching, graph construction, triangulation, and bundle adjustment to generate a geometrically consistent sparse point cloud. The outputs serve as physically meaningful initialization for 3DGS, mitigating floating primitives and unstable training in the original EOGS framework.

[46] have proposed locally approximating RPC-based imaging relationships with equivalent perspective camera models, followed by the application of standard SfM pipelines for camera parameter estimation and sparse point cloud reconstruction (Fig. 3). We adapt its core idea to the EOGS framework [31]. This initialization provides a physically meaningful geometric support for subsequent Gaussian optimization, without altering the original EOGS formulation. The detailed implementation is described as follows.

1) Satellite-Adapted SfM for Sparse Geometry Recovery: Due to the large sensor-to-ground distance and limited depth variation within the AOI, the RPC model can be locally approximated by a weak-perspective camera model. Based on the above formulation, the RPC model can be approximated by an equivalent perspective camera model. Specifically, a set

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of 3-D points X_i is sampled within the AOI bounding volume

$$\left\{X_i = (x_i, y_i, z_i)^\top\right\} \tag{1}$$

and their corresponding projections on the image plane (u_i, v_i) are obtained via the RPC model function

$$(u_i, v_i) = \mathrm{RPC}\,(X_i)\,. \tag{2}$$

Given these 3-D–2-D correspondences, the projection matrix P is estimated using direct linear transformation (DLT)

$$\lambda_i \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = P \begin{bmatrix} x_i \\ y_i \\ z_i \\ 1 \end{bmatrix} \tag{3}$$

where λ_i denotes the homogeneous scale factor. The projection matrix is then decomposed as

$$P = K\,[R \mid t] \tag{4}$$

from which the intrinsic matrix K and extrinsic parameters (R, t) of the equivalent perspective camera are obtained.

During multiview geometric reconstruction, considering the significant illumination variations and limited observation baselines in satellite imagery, the scale-invariant feature transform (SIFT) is employed for cross-image matching. Reliable correspondences are further selected by enforcing geometric consistency constraints to filter out outliers. Based on the refined matches, a sparse 3-D point cloud is reconstructed via triangulation. Finally, bundle adjustment is performed by minimizing the global reprojection error to jointly optimize the camera parameters and 3-D points, thereby further improving reconstruction accuracy.

2) Coordinate Normalization and Scene Mapping: The sparse point cloud recovered by the aforementioned SfM process is represented in the local east-north-up (ENU) coordinate system and cannot be directly used in the internal scene representation of EOGS. Therefore, it is necessary to transform it into a unified normalized world coordinate system. This transformation involves four sequential steps: ENU → Earth-centered Earth-fixed (ECEF) → geographic coordinates → UTM → normalized coordinates.

Let a point in the ENU coordinate system be denoted as

$$\boldsymbol{X}_i^{\mathrm{enu}} = \begin{bmatrix} e_i \\ n_i \\ u_i \end{bmatrix}. \tag{5}$$

Its conversion to the ECEF coordinate system is expressed as

$$\boldsymbol{X}_i^{\mathrm{ecef}} = R_{\mathrm{enu}\to\mathrm{ecef}} \cdot \boldsymbol{X}_i^{\mathrm{enu}} + \boldsymbol{X}_0^{\mathrm{ecef}} \tag{6}$$

where $\boldsymbol{X}_0^{\mathrm{ecef}}$ denotes the position of the ENU origin in the ECEF system, corresponding to geographic coordinates (ϕ_0, λ_0, h_0).

The rotation matrix is defined as

$$R_{\mathrm{enu}\to\mathrm{ecef}} = \begin{bmatrix} -\sin\lambda_0 & -\sin\phi_0\cos\lambda_0 & \cos\phi_0\cos\lambda_0 \\ \cos\lambda_0 & -\sin\phi_0\sin\lambda_0 & \cos\phi_0\sin\lambda_0 \\ 0 & \cos\phi_0 & \sin\phi_0 \end{bmatrix}. \tag{7}$$

Subsequently, the ECEF coordinates are converted to geographic coordinates

$$\boldsymbol{X}_i^{\mathrm{ecef}} \to (\lambda_i, \phi_i, h_i) \tag{8}$$

and further projected to the UTM plane coordinates

$$(\lambda_i, \phi_i) \to \left(x_i^{\mathrm{utm}}, y_i^{\mathrm{utm}}\right). \tag{9}$$

Finally, the UTM coordinates are affine-normalized to obtain the world coordinates used within EOGS

$$\boldsymbol{X}_i^{\mathrm{NDC}} = \frac{\boldsymbol{X}_i^{\mathrm{utm}} - \boldsymbol{c}}{s} \tag{10}$$

where $\boldsymbol{c}$ and s denote the scene center and scale factor, respectively. The overall transformation can be summarized as

$$\boldsymbol{X}_i^{\mathrm{NDC}} = \frac{\Pi_{\mathrm{utm}}\left(\Pi_{\mathrm{geo}}\left(R_{\mathrm{enu}\to\mathrm{ecef}} \cdot \boldsymbol{X}_i^{\mathrm{enu}} + \boldsymbol{X}_0^{\mathrm{ecef}}\right)\right) - \boldsymbol{c}}{s}. \tag{11}$$

3) Gaussian Primitive Initialization: The method employs SfM to recover an initial sparse point cloud from multiview images, which is subsequently converted into a set of parameterized 3-D Gaussian ellipsoids $G(\boldsymbol{X}_i^{\mathrm{NDC}})$. Each Gaussian primitive is precisely characterized by its spatial position, covariance matrix, opacity, and spherical harmonic coefficients. The 3-D Gaussian function is defined as

$$G\left(\boldsymbol{x}_i^{\mathrm{NDC}}\right) = \frac{1}{\sqrt{(2\pi)^k\,|\boldsymbol{\Sigma}|}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})} \tag{12}$$

where $\boldsymbol{X}_i^{\mathrm{NDC}} \in \mathbb{R}^3$ denotes an arbitrary point in NDC, and $\boldsymbol{\mu} \in \mathbb{R}^3$ represents the center of the 3-D Gaussian, indicating the spatial location of the Gaussian primitive in the world coordinate system. k is the dimension of the Gaussian distribution, which is fixed to $k = 3$ for 3-D Gaussians, corresponding to the 3-D spatial coordinates (x, y, z). The anisotropic covariance matrix $\boldsymbol{\Sigma} = \boldsymbol{A}\boldsymbol{A}^T$ defines the shape of the 3-D Gaussian, where $\boldsymbol{A} = \boldsymbol{R}\boldsymbol{S}$, with $\boldsymbol{R}$ being a rotation matrix, and $\boldsymbol{S}$ being a scaling matrix. Through the linear transformation matrix $\boldsymbol{A}$, a standard isotropic Gaussian can be transformed into an arbitrarily shaped Gaussian ellipsoid. For the ith point p_i, the center of the corresponding Gaussian primitive is initialized as

$$\mu_i = p_i. \tag{13}$$

The covariance matrix is parameterized in a decomposable form

$$\Sigma_i = R_i S_i S_i^T R_i^T \tag{14}$$

where R_i denotes the rotation matrix, and $S_i = \mathrm{diag}(s_{i,1}, s_{i,2}, s_{i,3})$ is a diagonal scale matrix. During the initialization stage, for numerical stability, we set

$$R_i = I \tag{15}$$

resulting in an isotropic initial Gaussian distribution.

The scale parameters S_i are adaptively estimated based on the local spatial distribution of the point cloud. Specifically, the scale for each point is computed as the mean Euclidean distance between p_i and its K nearest neighbors

$$s_i = \frac{1}{K} \sum_{j \in \mathcal{N}_K(i)} \left\| p_i - p_j \right\|_2 \tag{16}$$

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where $\mathcal{N}_K(i)$ denotes the set of indices of the K nearest neighbors of p_i, and $\|\cdot\|_2$ represents the L_2 norm (Euclidean distance) in 3-D space. This allows the Gaussian primitives to adapt to the local point density and better capture the spatial structure of the scene.

In addition, the opacity parameter of each 3-D Gaussian is parameterized using the Sigmoid function to ensure it is constrained within the valid range $[0, 1]$

$$\alpha_i = \sigma(\widetilde{\alpha}_i) = \frac{1}{1 + e^{-\widetilde{\alpha}_i}} \tag{17}$$

where α_i denotes the final opacity of the ith Gaussian, $\widetilde{\alpha}_i$ is the unconstrained learnable preactivation parameter optimized during training, and $\sigma(\cdot)$ represents the Sigmoid activation function.

C. VGG-Based Perceptual Loss Module

To further improve the visual quality and fine-grained texture fidelity of orthographic rendering results, we incorporate a VGG-based perceptual loss [57] as an auxiliary supervision term on top of the original multilevel loss formulation. The underlying 3DGS framework already integrates several advanced objectives, including pixel-wise L1 loss, structural similarity loss, multiview geometric consistency constraints, and terrain smoothness regularization, forming a comprehensive optimization scheme from pixel appearance to geometric structure. However, in satellite imagery, certain ground regions are persistently affected by shadows or low illumination conditions due to limited solar angles and acquisition geometry. These regions often exhibit insufficient texture variation across views, making them poorly constrained by pixel-level consistency losses alone and prone to over-smoothing or local artifacts in the rendered orthographic images. To alleviate this issue, we introduce a perceptual loss computed in the feature space of a pretrained VGG network. By enforcing consistency between high-level feature representations of the rendered image and the reference image, this loss provides a global appearance prior that complements pixel-based supervision. Although applied over the entire image, the perceptual constraint primarily contributes to regions with weak pixel-level signals, such as shadowed areas, improving texture coherence and visual completeness. The perceptual loss is defined as

$$\mathcal{L}_{\text{perc}} = \sum_l \lambda_l \left| \phi_l\left(\hat{I}\right) - \phi_l(I) \right| \tag{18}$$

where $\phi_l(\cdot)$ denotes the feature extraction function at the lth layer of the VGG network, $\hat{I}$ and I represent the rendered and reference orthographic images, respectively, and λ_l balances the contributions of different feature levels.

In practice, we employ a VGG-19 network and extract perceptual features from layers conv1_2, conv2_2, conv3_3, and conv4_3. The corresponding weights λ_l are empirically tuned to balance low-level texture details and higher level structural patterns. Notably, since the perceptual loss operates solely on the rendered images and does not directly modify the underlying 3-D Gaussian geometry, it enhances appearance modeling without compromising geometric consistency.

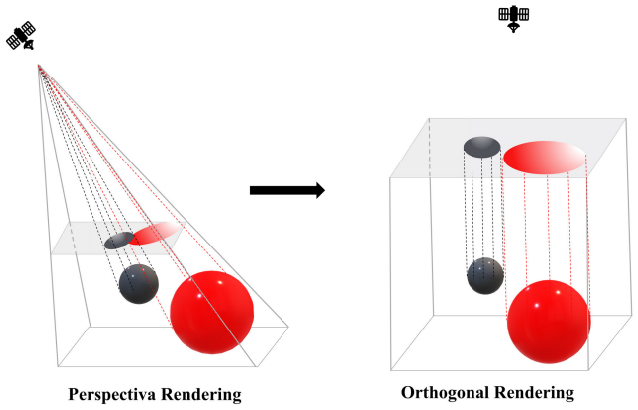


Fig. 4. Illustration of projection frustums. (Left) Perspective projection frustum, where object appearance follows the near-large–far-small principle. (Right) Orthographic projection frustum, in which object imaging is independent.

The final optimization objective is formulated as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_z + \alpha \mathcal{L}_{\text{perc}} \tag{19}$$

where $\mathcal{L}_z$ denotes the original regularization and reconstruction loss, and α controls the relative importance of the perceptual term. Through joint optimization, this strategy effectively improves the subjective visual quality of the generated orthographic images, yielding sharper boundaries, more coherent textures, and improved color consistency.

D. Satellite Gaussian Orthographic Rendering Module

To adapt 3DGS to orthographic projection, we propose a novel satellite-oriented true orthographic rasterization rendering module, as illustrated in Fig. 4. Specifically, we construct an equivalent nadir-view virtual camera based on the original satellite camera poses to eliminate perspective components induced by off-nadir imaging, and define a parallel viewing frustum that satisfies orthographic imaging constraints.

1) Virtual Camera Construction: To enable orthographic rendering within the 3DGS framework and remove perspective distortion caused by camera tilt, we first define an orthographic projection camera. Given world-to-camera transformation matrix T_{orig}, we decompose it into a rotation matrix T_{orig} and a translation vector $\boldsymbol{b}$

$$T_{\text{orig}} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{b} \\ \boldsymbol{0}^T & 1 \end{bmatrix}. \tag{20}$$

We define a canonical direction vector $\boldsymbol{d} = [0, 0, 1]^T$, which corresponds to the nadir viewing direction in camera coordinates. This vector is projected into the current camera coordinate system to obtain the planar offset of the orthographic projection direction on the image plane

$$\boldsymbol{q} = \boldsymbol{A}\boldsymbol{d} = \left[q_x, q_y, 1\right]^T. \tag{21}$$

To compensate for this offset, we construct a correction matrix

$$\boldsymbol{M} = \begin{bmatrix} 1 & 0 & -q_x \\ 0 & 1 & -q_y \\ 0 & 0 & 1 \end{bmatrix}$$

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which applies translational compensation along the image-plane x- and y-directions. Finally, by incorporating the scene center $\boldsymbol{c}$, we compute the updated rotation and translation parameters and assemble the new transformation matrix for the orthographic camera

$$T_{\text{nadir}} = \begin{bmatrix} \boldsymbol{MA} & (\boldsymbol{I}-\boldsymbol{M})\boldsymbol{Ac}+\boldsymbol{b} \\ \boldsymbol{0}^T & 1 \end{bmatrix}. \tag{22}$$

This virtual camera is used for orthographic rendering, ensuring that the output image is strictly aligned with the ground plane and free from geometric distortions caused by the original off-nadir viewing geometry.

2) Parallel Frustum Construction: After determining the virtual camera pose, we further filter Gaussian ellipsoids that fall within the camera's viewing volume to exclude irrelevant primitives and improve rendering efficiency. In contrast to the conventional pyramidal view frustum used in perspective projection, the orthographic rendering pipeline adopts a prismatic parallel frustum, which is consistent with the geometric constraints of orthographic imaging.

3) Orthographic Rasterization Rendering: The core of the proposed approach lies in an orthographic rasterization module specifically designed for TDOM generation within the 3DGS framework. The original 3DGS pipeline relies on perspective projection, which introduces depth-dependent scaling effects. In contrast, we reformulate the Gaussian projection process under an orthographic imaging model. Specifically, the perspective projection matrix used in the Gaussian covariance transformation is replaced with an orthographic projection matrix, ensuring that all projection rays remain parallel and perpendicular to the image plane. As a result, the projected footprint of each Gaussian kernel on the image plane depends solely on its planar spatial location, rather than its distance from the camera center, thereby effectively eliminating geometric distortions caused by perspective effects. Formally, the covariance transformation of an individual 3-D Gaussian kernel is expressed as

$$\boldsymbol{\Sigma}_k' = \boldsymbol{K}\boldsymbol{\Sigma}_k\boldsymbol{K}^T \tag{23}$$

where $\boldsymbol{K}$ denotes the orthographic projection matrix

$$\boldsymbol{K} = \begin{bmatrix} \frac{2}{w} & 0 & 0 & 0 \\ 0 & \frac{2}{h} & 0 & 0 \\ 0 & 0 & -\frac{2}{d} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

with w, h, and d representing the width, height, and depth of the orthographic viewing volume, respectively. During differentiable rendering, each 3-D Gaussian primitive is projected onto the image plane via a differentiable transformation to form an anisotropic 2-D Gaussian distribution.

Following the standard 3DGS formulation, the covariance transformation is given by:

$$\boldsymbol{\Sigma}' = \boldsymbol{JW}\boldsymbol{\Sigma}_k'\boldsymbol{W}^T\boldsymbol{J}^T \tag{24}$$

where $\boldsymbol{J}$ denotes the Jacobian of the projection function, and $\boldsymbol{W}$ represents the world-to-camera transformation matrix.



Fig. 5. DFC2019 dataset, including RGB images from (a) JAX_004, (b) JAX_068, (c) JAX_214, and (d) JAX_260 scenes.

Finally, using a tiled rasterization strategy, overlapping Gaussian distributions are sorted and accumulated at each pixel location via alpha blending to generate the rendered image

$$C = \sum_{i\in N} c_i\alpha_i T_i. \tag{25}$$

With

$$T_i = \prod_{j=1}^{i-1}(1-\alpha_j) \tag{26}$$

where N is the set of 3-D Gaussians covering the current pixel; c_i and α_i are the color and opacity of the ith Gaussian, respectively; α_j is the opacity of the jth Gaussian in front of the ith one; and T_i denotes the transmittance of the ith Gaussian, reflecting its remaining visibility after occlusion by preceding Gaussians.

IV. Experiments

A. Datasets

To comprehensively evaluate the effectiveness and generalization capability of the proposed method, experiments are conducted on two publicly available multiview satellite datasets: the IEEE GRSS Data Fusion Contest 2019 (DFC2019) dataset and the MVS3DM dataset. These datasets cover diverse geographic regions, urban structures, and imaging conditions, providing a solid basis for evaluating both geometric accuracy and visual quality in TDOM generation.

The DFC2019 dataset originates from the 2019 IEEE GRSS Data Fusion Contest, covering the Jacksonville area, Florida, USA (JAX). It contains multiview optical imagery acquired by the WorldView-3 satellite with a ground sampling distance (GSD) of approximately 0.3 m, along with

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TABLE I
SUMMARY OF THE AOIS USED IN THE EXPERIMENTS FROM THE DFC2019 AND MVS3DM DATASETS

Dataset	Scene	ImageNum	Min_alt	Max_alt	Lat	Lon	Scene Characteristics
DFC2019	JAX_004	11	-24	1	30.357	-81.706	Suburban residential area with waterbody
	JAX_068	19	-27	30	30.348	-81.663	Mixed urban zone (moderate density)
	JAX_214	24	-29	73	30.316	-81.663	Dense urban core (high-rise)
	JAX_260	17	-30	13	30.311	-81.663	Waterfront urban area
MVS3DM	IARPA_001	29	12	59	-34.490	-58.584	High-density urban residential district
	IARPA_002	24	12	80	-34.447	-58.575	Planned urban community
	IARPA_003	23	12	80	-34.418	-58.575	Industrial/commercial complex with waterbody

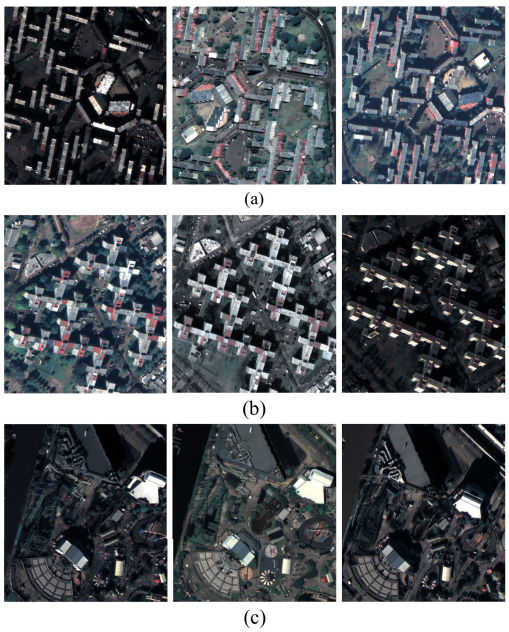


Fig. 6. MVS3DM dataset, including raw pansharpened images from (a) IARPA_001, (b) IARPA_002, and (c) IARPA_003 scenes.

high-resolution LiDAR-derived DSMs (30–50 cm/pixel) as reference for geometric evaluation (see Fig. 5). The MVS3DM dataset is sourced from the IARPA Multi-View Stereo (MVS) 3-D Mapping Challenge, covering the San Fernando region in Argentina (IARPA). It includes multiview panchromatic (GSD 0.31 m) and multispectral imagery (GSD 1.24 m), together with airborne LiDAR point clouds that provide precise elevation references (see Fig. 6).

Seven representative AOIs were selected from these datasets, four from DFC2019 (JAX_004, JAX_068, JAX_214, JAX_260) and three from MVS3DM (IARPA_001, IARPA_002, IARPA_003). The selected AOIs cover a wide range of representative satellite imaging conditions and urban scenarios. The chosen AOIs include densely built urban cores with severe occlusions and complex building structures, as well as relatively open suburban and waterfront regions with diverse terrain and land-cover characteristics. In addition, the datasets exhibit varied imaging conditions, including different viewing geometries, illumination patterns, and sensor modalities. Such diversity enables a comprehensive evaluation of the proposed method in terms of geometric accuracy, occlusion robustness, radiometric consistency, and generalization capability across realistic satellite TDOM generation scenarios. Each AOI contains 11–29 images captured from different viewpoints. The selected AOIs cover diverse urban morphologies and land cover types, including low-density suburban areas, high-rise urban cores, planned residential communities, waterfront regions, and industrial-mixed zones (see Figs. 5 and 6). These scenes vary substantially in building density, terrain relief, and water coverage, enabling comprehensive assessment of the proposed method's robustness across different urban contexts. For computational efficiency and scene completeness, all AOIs were cropped to approximately 800×800 pixels. Detailed information about the AOIs is summarized in Table I. LiDAR DSMs with a spatial resolution ranging from 30 to 50 cm per pixel are used as reference data for elevation evaluation. The quality of the generated results is assessed using image-based metrics, including peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), and learned perceptual image patch similarity (LPIPS) [58], as well as the mean absolute error (MAE) of elevation with respect to the LiDAR DSM.

B. Details

The proposed Tortho–SatGS method is implemented based on the EOGS code base. For Gaussian primitive parameters, we adopt the original EOGS optimizer and learning rate scheduler. Camera-related parameters are optimized using a second Adam optimizer with a learning rate of 10^{-2}, which is used to learn camera-dependent parameters for affine color correction and ambient light color. The total number of training iterations is set to 30 000. Shadow mapping and the original regularization terms are activated after 1000 iterations, while the VGG-based perceptual loss is introduced at iteration 2000. The Lagrange multiplier for the perceptual loss is selected through validation on a single scene and then applied to all other scenes, consistently yielding stable performance. The final perceptual loss weight is set to 0.01, demonstrating the robustness of the model with respect to this parameter. In addition, the sparse point cloud generated by satellite SfM is used to replace the uniform initialization of Gaussian centers, and the initial Gaussian scale is adaptively estimated from local neighborhood distances, with an average value of approximately 0.025. In the rendering pipeline, an additional nadir-view rendering configuration is introduced,

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and an orthographic projection matrix is employed to generate the final true orthophoto images.

We compare our method with three representative baselines: 1) high-based: a traditional orthorectification pipeline based on Shuttle Radar Topography Mission (SRTM)-derived elevation data; 2) a NeRF-based neural rendering method adapted for satellite TDOM generation; and 3) the commercial photogrammetry software Metashape. These baselines, respectively, represent geometry-driven, implicit neural, and industrial solutions.

All compared methods are trained and evaluated using the same input imagery and camera parameters to ensure a fair comparison. For NeRF-based methods, satellite camera modeling and orthographic rendering adaptations are applied to align with the TDOM generation setting. All experiments are conducted on a workstation equipped with an Intel[1] Core i5-12490F CPU and an NVIDIA RTX 3050 GPU, with GPU acceleration enabled.

C. Evaluation Metrics

To comprehensively evaluate both geometric accuracy and visual quality, multiple metrics are adopted. Specifically, MAE is used as the primary metric to assess geometric consistency by comparing the reconstructed elevation with reference data. In addition, PSNR and SSIM are used to evaluate pixel-level reconstruction quality, while LPIPS measures perceptual similarity. The no-reference blind/referenceless image spatial quality evaluator (BRISQUE) metric is further introduced to assess perceptual quality without ground truth.

1) Mean Absolute Error: MAE is used to measure the elevation discrepancy between the predicted DSM and the reference DSM

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i| \tag{27}$$

where $\hat{y}_i$ denotes the predicted DSM value at pixel i, y_i is the corresponding reference DSM value, and N is the number of valid pixels involved in the computation.

2) Model Reconstruction Accuracy: The credibility of a TDOM fundamentally depends on its geometric fidelity, i.e., the extent to which the generated image accurately maps the true spatial locations and shapes of ground objects. This property is directly related to the accuracy of the underlying 3-D scene reconstruction. Following the evaluation protocol of Ortho-3DGS [52], we adopt PSNR, SSIM, and LPIPS as indirect indicators to quantify the reconstruction quality of the 3DGS-based scene model, thereby reflecting the reliability of the generated TDOM.

Peak Signal-to-Noise Ratio: PSNR measures the overall image quality and its similarity to the reference image

$$\mathrm{PSNR} = 10 \cdot \log_{10}\frac{\mathrm{MAX}^2}{\mathrm{mse}} \tag{28}$$

where *mse* denotes the mean squared error, and *MAX* is the maximum possible pixel value.

Structural Similarity Index Measure: SSIM evaluates image similarity in terms of luminance, contrast, and structural information

$$\mathrm{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}. \tag{29}$$

Learned Perceptual Image Patch Similarity: LPIPS measures perceptual similarity between two images using deep neural network features: LPIPS [58]

$$\mathrm{LPIPS}(x,y) = \sum_i w_i \cdot \|\phi_i(x) - \phi_i(y)\|_2^2 \tag{30}$$

where ϕ_i denotes feature maps from the ith layer of a pretrained deep network, and w_i is learned weights.

3) No-Reference Image Quality Assessment (BRISQUE): In addition to full-reference metrics, we further employ BRISQUE to evaluate the naturalness, radiometric consistency, and local artifact level of the generated true orthophotos. As a no-reference image quality metric, BRISQUE estimates image quality by measuring the deviation of an image from natural scene statistics, which is particularly useful in shadowed regions, low-illumination areas, and weak-texture water surfaces where full-reference metrics may be less informative. It can be formulated as

$$\mathrm{BRISQUE}(I) = \mathrm{SVR}(\Phi(\mathrm{MSCN}(I))) \tag{31}$$

where I denotes the image to be evaluated, MSCN(I) represents the mean-subtracted contrast-normalized coefficients, $\Phi(\cdot)$ denotes the feature vector extracted based on an asymmetric generalized Gaussian distribution, and SVR is a pretrained support vector regression model. A lower BRISQUE score indicates better perceptual quality.

Among these metrics, MAE is considered the most critical indicator for TDOM generation, as it directly reflects geometric correctness and adherence to orthographic projection. The other metrics provide complementary evaluation from visual and perceptual perspectives.

D. Comparative Experiment

1) Efficiency: We further analyze the computational efficiency of the proposed Tortho–SatGS method in terms of training cost, rendering efficiency, memory footprint, and scalability. All experiments are conducted on a workstation equipped with an NVIDIA RTX 3050 GPU. The total number of training iterations is set to 30 000.

a) Training efficiency: As shown in Table II, the proposed SfM-based initialization significantly improves training efficiency. Compared with the original EOGS, the training time is reduced from 0.80–1.05 h to approximately 0.25–0.38 h across different scenes. This improvement is mainly attributed to the substantial reduction in the number of Gaussian primitives, which decreases optimization complexity and accelerates convergence.

Compared with NeRF-based methods, the efficiency advantage is more pronounced. NeRF-based approaches require an average training time of 9.76 h, whereas the proposed method achieves an average training time of only 0.36 h, corresponding to approximately 27× faster training. This gain

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TABLE II
COMPUTATIONAL EFFICIENCY AND MEMORY FOOTPRINT COMPARISON AMONG THE PROPOSED METHOD, EOGS, AND NERF-BASED APPROACHES

	Method	JAX_004	JAX_068	JAX_214	JAX_260	IARPA_001	IARPA_002	IARPA_003
Training (h)	NeRF-based	9.75	9.20	9.57	13.63	9.76	9.20	7.23
	EOGS	0.82	0.99	1.05	0.9	0.8	0.9	0.88
	ours	0.33	0.38	0.39	0.37	0.33	0.33	0.38
Rendering (s)	NeRF-based	12	12	14	11	11	13	12
	EOGS	0.7	0.8	1.2	1.0	0.8	0.7	0.7
	ours	0.7	0.9	1.2	1.0	0.7	0.6	0.5
Memory (GB)	NeRF-based	5.1	5.1	5.0	5.0	5.1	5.1	5.1
	EOGS	7.9	9.2	8.9	6.8	8.1	20.7	7.1
	ours	3.7	3.4	3.5	3.4	3.6	3.8	3.4

mainly stems from avoiding ray-based volumetric integration and repeated MLP evaluations, enabling more efficient optimization based on explicit geometric primitives.

b) Rendering efficiency: The computational complexity of NeRF-based rendering can be approximated as $O(N_{\text{rays}} \times N_{\text{samples}} \times C_{\text{MLP}})$, whereas the proposed method reduces this to $O(N_{\text{visible_gaussians}})$ through efficient rasterization and visibility culling.

As a result, NeRF-based methods require an average rendering time of 12.1 s per orthophoto, while our method achieves approximately 0.8 s, corresponding to about 15× faster inference. This efficiency advantage arises from two key design choices. First, the proposed method eliminates ray-based volumetric sampling, thereby avoiding costly ray marching and repeated neural network inference. Second, it adopts a GPU rasterization-based 3DGS representation, allowing direct projection of primitives and significantly accelerating image synthesis.

c) Memory consumption: As shown in Table II, the proposed method achieves the lowest memory usage across all scenes, compared to approximately 5.0–5.1 GB for NeRF-based methods and significantly higher, more variable consumption for EOGS.

This improvement mainly results from the substantial reduction in the number of Gaussian primitives enabled by SfM-based initialization. As shown in the ablation study, the number of primitives decreases from the million level in EOGS to approximately 10^4, leading to a much more compact representation.

In addition, the rasterization-based rendering pipeline avoids storing per-ray sampled features required by NeRF, and processes only visible Gaussian primitives, further improving memory efficiency.

Overall, the proposed method provides a compact and memory-efficient solution for stable training and inference across diverse satellite scenes.

d) Scalability analysis: To evaluate scalability with respect to the number of input views, we conduct experiments using 25%, 50%, and 100% of the available images, as shown in Fig. 7.

As the number of input images increases, the training time grows moderately, indicating stable scalability. Meanwhile, the MAE is significantly reduced, demonstrating that additional views provide stronger multiview geometric constraints and improve reconstruction accuracy. In contrast, SSIM shows a slight decrease, which can be interpreted as a transition from single-view pixel fitting to globally consistent 3-D

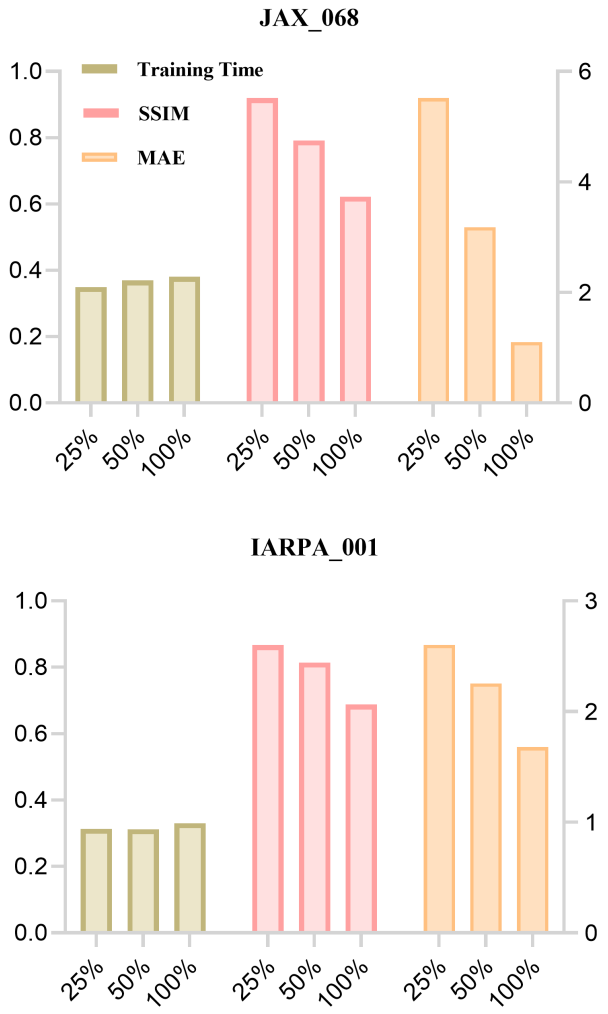


Fig. 7. Performance metrics (training time, SSIM, MAE) of the proposed Tortho–SatGS method on the JAX_068 and IARPA_001 scenes, evaluated with 25%, 50%, and 100% of the available input images.

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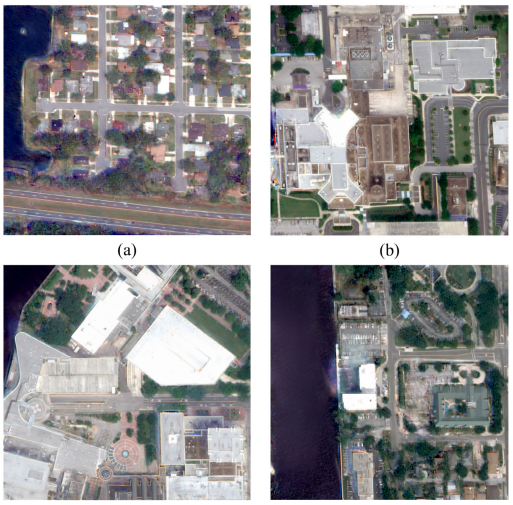


Fig. 8. Orthophoto results of Tortho–SatGS on the DFC2019 dataset. (a) JAX_004. (b) JAX_068. (c) JAX_214. (d) JAX_260.



Fig. 9. Orthophoto results of Tortho–SatGS on the MVS3DM dataset. (a) IARPA_001. (b) IARPA_002. (c) IARPA_003.

optimization. This reflects a tradeoff where improved geometric consistency may slightly reduce pixel-wise similarity but leads to more physically accurate reconstructions.

Overall, the proposed Tortho–SatGS achieves a favorable balance between efficiency and reconstruction quality, making it suitable for large-scale satellite orthophoto generation tasks.

2) Qualitative Analysis: In the qualitative evaluation, we focus on two fundamental and complementary aspects of TDOM generation: geometric accuracy and radiometric characteristics. Together, these two factors determine the spatial usability and visual reliability of the generated orthophoto imagery.

a) Geometric analysis: Figs. 8 and 9 present the TDOM generation results of Tortho–SatGS on the DFC2019 and MVS3DM datasets, respectively. It can be observed that, across diverse urban scenes and land-cover types, the generated TDOM maintains strong geometric consistency along building boundaries. Common artifacts associated with perspective imaging—such as residual building facades, edge bending, and projection distortions—are effectively suppressed. These observations indicate that Tortho–SatGS can stably generate high-quality orthophotos that conform to orthographic geometric constraints across complex and heterogeneous scenes.

As illustrated in Fig. 10, different methods exhibit notable differences in geometric consistency, structural integrity, and texture stability. The traditional Height-based method is able to preserve relatively high spatial resolution in local regions; however, due to its reliance on single-view imagery or simplified geometric correction models, it fails to fully eliminate projection distortions induced by oblique viewing angles. In building-dense areas, roof structures often suffer from varying degrees of tilt-induced stretching or boundary displacement, which degrades TDOM geometric accuracy. NeRF-based methods alleviate perspective-induced structural distortions to some extent through implicit volumetric rendering, resulting in more regular overall building shapes. Nevertheless, due to implicit continuous scene modeling and limited sampling density during volume rendering, high-frequency textures tend to be oversmoothed, leading to insufficient detail representation. As a mature SfM/MVS reconstruction tool, Metashape can recover overall object geometry reasonably well; however, its orthorectification results heavily depend on point cloud density and image fusion strategies. In complex urban environments, artifacts such as seam lines, local blurring, or slight roof tilting may still occur, and color consistency becomes difficult to maintain under multitemporal or varying illumination conditions.

A more detailed analysis of representative scenes is shown in Fig. 11. In building-dominated areas [Fig. 11(a)], both the height-based method and Metashape exhibit noticeable geometric deformation and boundary blurring near roof edges and facade transition regions. While the NeRF-based approach preserves overall structural integrity, it shows instability in shadowed regions and weak texture consistency across viewpoints. In contrast, Tortho–SatGS produces roof structures with clear contours and coherent geometry, effectively avoiding facade residues and edge bending, while maintaining stable, noise-free textures in shadowed areas. This demonstrates stronger multiview geometric consistency and illumination robustness. For fine-scale structures [Fig. 11(b)], NeRF-based methods tend to lose high-frequency details due to hierarchical sampling and implicit smoothing in volumetric rendering, resulting in texture over-blending or blurred boundaries. By contrast, Tortho–SatGS preserves orthographic geometric constraints while retaining sharper building edges and fine structural details, yielding clearer boundaries and higher local contrast. This highlights its advantage in recovering high-frequency geometric and texture cues. Water surfaces present additional challenges due to their dynamic appearance and lack of stable texture [Fig. 11(c)]. The height-based method suffers from color discontinuities and block-like artifacts, while NeRF-based methods introduce severe color drift and view-dependent artifacts. Metashape produces smoother surfaces but lacks sufficient detail and natural reflection transitions. Our method generates the most stable and uniform water appearance, with consistent color distribution and accurate boundary transitions, demonstrating superior handling of low-texture regions. Vegetation scenes [Fig. 11(d)] involve complex geometry and nonrigid structures with high texture variability. NeRF-based methods exhibit significant detail loss, with tree canopies appearing overly blurred. Metashape preserves coarse geometric shapes but lacks fine-grained texture detail. In contrast, Tortho–SatGS achieves clearer canopy textures, more

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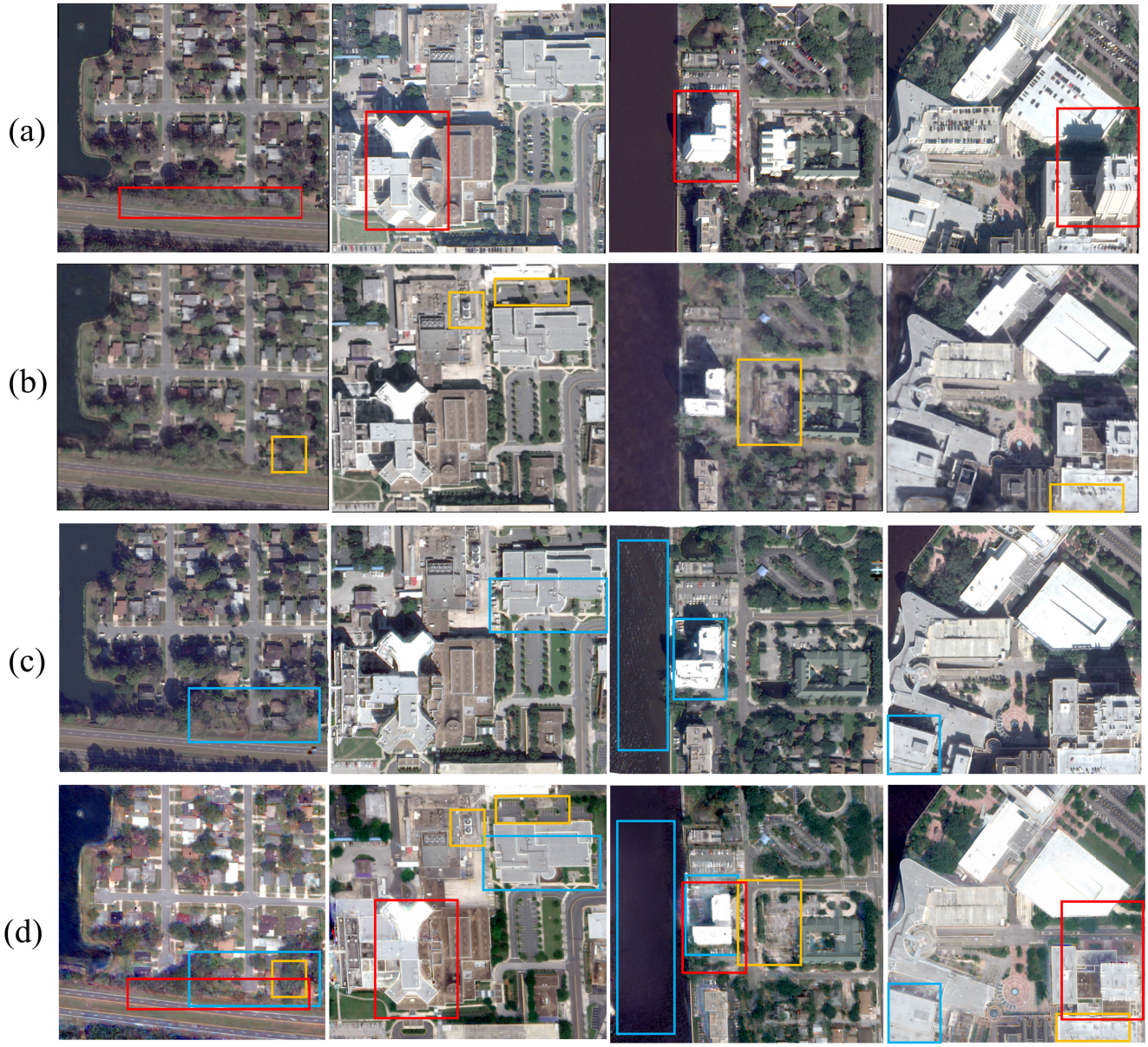


Fig. 10. Comparison of orthophotos generated by different methods. (a)–(d) Results obtained by the high-based, NeRF-based, Metashape, and the proposed method, respectively. The proposed method achieves superior performance in building regions, fine-scale structures, water surfaces, and vegetation areas. Red boxes highlight regions of interest.

distinct boundaries, and more natural shadow distributions, indicating improved performance in such complex scenarios.

b) Radiation characteristics: High-quality natural images typically follow specific statistical distributions, which are disrupted by distortions. To analyze radiometric characteristics, we adopt the no-reference image quality assessment metric BRISQUE. BRISQUE evaluates image quality by measuring deviations from natural scene statistical regularities.

As reported in Table III, our method achieves the most stable no-reference quality performance on the DFC2019 dataset. Although the three baseline methods demonstrate generally reliable performance—and NeRF-based methods occasionally achieve very high scores—they exhibit substantial variability across different scenes. In contrast, our model shows stronger robustness to scene variations, delivering competitive or superior results in challenging regions where other methods tend to degrade. These results confirm that the proposed approach provides more consistent reconstruction quality for satellite-based orthophoto generation. Overall, our method demonstrates clear advantages in both geometric and radiometric correction, producing orthophotos with accurate structures, balanced colors, and natural illumination. Geometric distortions and shadow-related errors caused by oblique viewing angles are largely eliminated. Nevertheless, slight brightness inconsistencies may still appear in locally complex illumination conditions or

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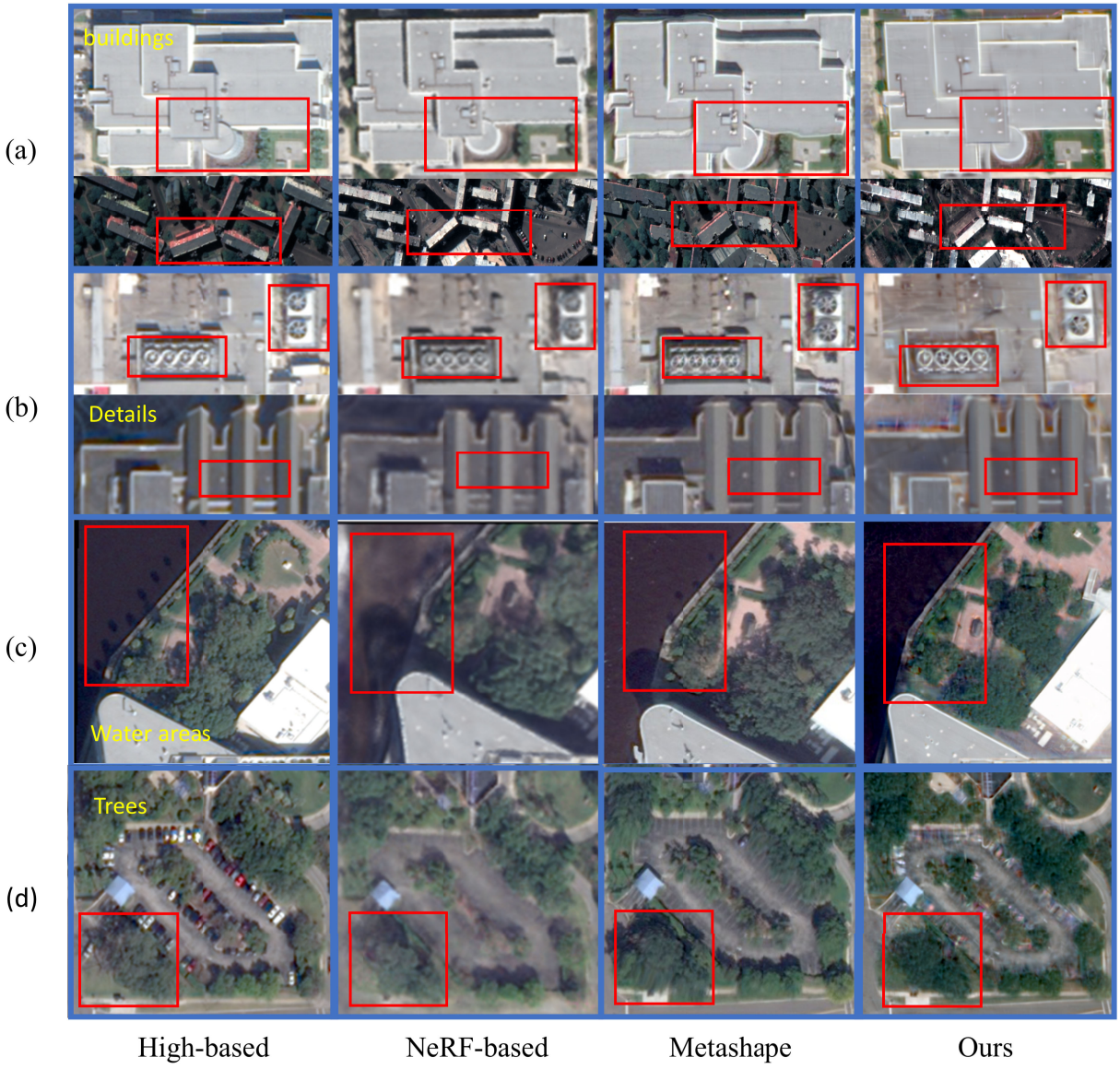


Fig. 11. Orthophoto outputs of the high-based, NeRF-based, Metashape, and the proposed Tortho–SatGS method, covering four typical scenarios (a) building regions, (b) fine-scale rooftop structures, (c) water areas, and (d) vegetation zones. Red boxes highlight key regions to demonstrate the advantages of the proposed method in geometric accuracy, detail preservation, and radiometric consistency.

TABLE III
BRISQUE PERFORMANCE COMPARISON FOR SATELLITE ORTHOPHOTO GENERATION

	BRISQUE			
	JAX_004	**JAX_068**	**JAX_214**	**JAX_260**
Truth	23.195	29.261	12.748	26.694
High-based	44.228	34.707	24.806	44.995
NeRF-based	28.414	51.549	33.561	26.374
Metashape	32.638	38.483	29.005	32.262
Ours	32.499	38.731	24.667	26.892

near object boundaries, primarily due to residual discrepancies between local reflectance properties and illumination estimation. In general, the limitations of the compared methods stem from insufficient modeling of true 3-D geometry and illumination relationships. By explicitly incorporating geometric constraints and color consistency optimization, our method significantly improves both geometric accuracy and radiometric consistency in true orthophoto generation.

3) Quantitative Analysis: We first evaluate the accuracy of surface reconstruction by computing the MAE between the DSMs generated from the reconstructed 3-D models and the reference DSMs derived from airborne LiDAR data, which serve as ground truth. Quantitative comparisons are conducted on two public datasets, MVS3DM and DFC2019. Using NeRF-based methods as the baseline, we adopt MAE as the evaluation metric and compare surface height reconstruction accuracy across seven representative scenes (IARPA_001,

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TABLE IV
MAE COMPARISON BETWEEN THE NERF-BASED METHOD AND THE PROPOSED METHOD ON THE MVS3DM AND DFC2019 DATASETS

Scene	IARPA_001	IARPA_002	IARPA_003	JAX_004	JAX_068	JAX_214	JAX_260
NeRF-based	3.3492	5.2426	4.9178	1.5292	1.3153	2.8620	2.4518
ours	1.6786	1.9671	1.8748	1.3873	1.1045	1.7206	1.5842

TABLE V
QUANTITATIVE MODEL EVALUATION RESULTS OF THE NERF-BASED METHOD AND THE PROPOSED METHOD ON THE DFC2019 DATASETS

Dataset / Method	JAX_004			JAX_068			JAX_214			JAX_260		
	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓
NeRF-based	0.879	26.415	0.127	0.912	25.21	0.141	0.949	25.234	0.079	0.839	21.767	0.119
Ours	0.793	26.44	0.323	0.622	25.398	0.276	0.825	21.888	0.315	0.705	24.068	0.345

IARPA_002, IARPA_003, JAX_004, JAX_068, JAX_214, and JAX_260). All results are averaged over multiple representative regions to ensure statistical reliability. The results are reported in Table IV.

Overall, our method consistently achieves lower MAE values than the NeRF-based approach across all seven scenes, indicating superior accuracy and robustness in DSM reconstruction. In particular, for the IARPA_001 scene, the NeRF-based Sat-NeRF framework lacks built-in radiometric normalization and real-time correction mechanisms for camera model errors, which prevents it from directly handling raw pansharpened products. As a result, it struggles to cope with color deviations, RPC inaccuracies, and complex radiometric noise present in the original data, leading to suboptimal reconstruction performance on the MVS3DM dataset. In contrast, our method attains substantially lower MAE values, demonstrating stable and consistent advantages. Similar performance gains are observed in the JAX-series scenes. Specifically, in JAX_004 and JAX_068, the MAE is reduced from 1.5292 and 1.3153 to 1.3873 and 1.1045, respectively. In more structurally complex scenes such as JAX_214 and JAX_260, the error reduction is even more pronounced, decreasing by approximately 1.14 and 0.87, respectively. These results suggest that the explicit point-based representation adopted by Tortho–SatGS provides more stable geometric fitting, particularly in urban environments characterized by dense high-rise buildings, severe occlusions, or large texture variations, where suppression of local reconstruction errors is especially critical.

Previous studies [11], [25] have demonstrated the feasibility of NeRF-based approaches for orthophoto generation from UAV and satellite imagery. Accordingly, we further evaluate the reconstructed models using three image quality metrics—SSIM, PSNR, and LPIPS—and compare the results with those obtained by NeRF-based methods. As shown in Tables V and VI, the quantitative evaluation of 3DGS-based scene reconstruction reflects both the geometric consistency and the visual fidelity of the rendered images.

On the MVS3DM dataset (Table VI), our method significantly outperforms the NeRF-based baseline across all three metrics, primarily due to the limitations of NeRF-based modeling in handling complex satellite imagery. On the DFC2019 dataset (Table V), performance exhibits certain scene-dependent characteristics. NeRF-based methods achieve relatively higher SSIM and LPIPS scores in some scenes, such as JAX_004 and JAX_214. In particular, for JAX_214, the NeRF-based reconstruction attains an SSIM of 0.949 and an LPIPS of 0.079, indicating that it can maintain reasonable structural consistency in mixed vegetation–building environments. By contrast, our method demonstrates superior PSNR performance in certain scenes, such as JAX_068, where a PSNR of 25.398 dB is achieved, slightly surpassing the NeRF-based result of 25.210 dB. This suggests competitive performance in noise suppression and global brightness restoration. Considering the combined results across both datasets, although different methods exhibit advantages under specific scenes or metrics, our approach consistently demonstrates competitive and stable performance across multiple evaluations. In particular, it achieves overall superior or comparable results in terms of SSIM and LPIPS relative to the NeRF-based baseline. These findings indicate that the Tortho–SatGS modeling strategy effectively preserves geometric consistency and visual fidelity, supporting its reliability for true orthophoto generation.

Although NeRF-based methods achieve higher SSIM in certain cases, SSIM mainly reflects structural similarity rather than true geometric correctness. Similarly, LPIPS focuses on perceptual similarity, which may favor smoother or visually consistent results. In contrast, our method prioritizes geometric fidelity and structural accuracy, which are more critical for TDOM generation.

Overall, the proposed method achieves a better balance between geometric accuracy and visual quality, making it more suitable for high-precision satellite orthophoto generation.

E. Ablation Experiment

1) Progressive Component Analysis: We conduct a series of ablation experiments to evaluate the contributions of the main design components in Tortho–SatGS, including the SfM-based Gaussian initialization, the orthographic rendering module, and the VGG-based perceptual supervision. The experiments are designed in a progressive manner to analyze how each

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TABLE VI
QUANTITATIVE MODEL EVALUATION RESULTS OF THE NERF-BASED METHOD AND THE PROPOSED METHOD ON THE MVS3DM DATASET

Dataset / Method	IARPA_001			IARPA_002			IARPA_003		
	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓
NeRF-based	0.327	11.863	0.719	0.260	10.673	0.673	0.375	15.145	0.575
Ours	0.688	21.413	0.344	0.685	23.63	0.356	0.705	24.065	0.345

TABLE VII
ABLATION STUDY RESULTS OF PROGRESSIVE COMPONENT INTEGRATION ON DFC2019 AND MVS3DM DATASETS

Scene	Method	Number of Gaussians	Runtime (h)	MAE ↓	PSNR ↑	SSIM ↑	LPIPS ↓
	EOGS	402597	0.82	3.0839	27.700	0.566	0.358
JAX_004	EOGS + SfM	15696	0.28	1.2857	26.130	0.806	0.419
	EOGS + SfM + VGG	15696	0.33	1.3873	26.440	0.793	0.323
	EOGS	1003939	0.99	1.8419	28.008	0.617	0.245
JAX_068	EOGS + SfM	23165	0.29	1.1171	25.211	0.661	0.422
	EOGS + SfM + VGG	23165	0.38	1.1045	25.398	0.622	0.276
	EOGS	2270522	1.05	2.4293	24.845	0.859	0.227
JAX_214	EOGS + SfM	25457	0.31	1.7113	21.695	0.832	0.452
	EOGS + SfM + VGG	25457	0.39	1.7204	21.888	0.825	0.315
	EOGS	673697	0.90	2.5316	26.474	0.874	0.335
JAX_260	EOGS + SfM	20400	0.25	1.5822	24.138	0.764	0.464
	EOGS + SfM + VGG	20400	0.37	1.5842	24.068	0.705	0.345
	EOGS	934292	0.80	2.2376	22.142	0.769	0.290
IARPA_001	EOGS + SfM	21751	0.29	1.6921	22.055	0.867	0.536
	EOGS + SfM + VGG	21751	0.33	1.6786	21.413	0.688	0.344
	EOGS	1473742	0.90	2.0788	23.687	0.677	0.356
IARPA_002	EOGS + SfM	13443	0.28	2.0209	23.747	0.632	0.358
	EOGS + SfM + VGG	13443	0.33	1.9671	23.630	0.685	0.356
	EOGS	1473536	0.88	2.0314	25.274	0.665	0.434
IARPA_003	EOGS + SfM	14412	0.27	1.8601	24.324	0.748	0.495
	EOGS + SfM + VGG	14412	0.38	1.8748	24.068	0.705	0.345

component contributes to geometric accuracy, rendering quality, and perceptual fidelity. The experiments are performed on seven representative scenes from the DFC2019 and MVS3DM datasets, covering diverse building structures, vegetation distributions, and radiometric imaging conditions. The analysis focuses on their effects on geometric accuracy, reconstruction quality, and perceptual fidelity.

The results (Table VII) show that introducing the SfM-based initialization substantially improves both geometric accuracy and computational efficiency. Compared with the original EOGS, the number of Gaussian primitives is reduced from the million level to approximately 1.3×10^4–2.5×10^4 across all scenes, while the training time decreases from 0.80–1.05 h to 0.25–0.31 h. More importantly, the MAE is consistently reduced in all seven scenes. These results indicate that the SfM-based sparse point cloud provides a physically meaningful geometric prior for Gaussian placement, thereby alleviating the instability caused by random initialization. As a result, the model achieves more accurate surface fitting with a much more compact representation. However, due to the substantial reduction in the number of Gaussian ellipsoids, the appearance modeling capacity is constrained to some extent, leading to scene-dependent fluctuations in PSNR and LPIPS.

Further introducing the VGG perceptual loss mainly improves the perceptual quality of the rendered orthophotos. Under the same sparse Gaussian initialization, the number of primitives remains unchanged, and the runtime increases only slightly. In contrast to the mixed behavior of PSNR and SSIM, LPIPS is consistently reduced in all scenes after adding the perceptual loss, indicating that the rendered results become perceptually closer to the reference images. This improvement can be attributed to the fact that the perceptual loss introduces feature-level supervision for regions with weak pixel-level constraints, such as shadowed areas, low-texture surfaces, and radiometrically inconsistent regions. Since the VGG term operates on the rendered images rather than directly constraining the underlying Gaussian geometry, its influence on MAE is limited, and the gains are mainly reflected in texture fidelity, color stability, and overall visual realism.

The visual results in Fig. 12 provide intuitive evidence of the progressive improvements introduced by each component. The original EOGS results exhibit noticeable geometric distortions, including blurred building boundaries, irregular rooftop shapes, and local stretching artifacts, indicating insufficient geometric constraints.

After introducing orthographic rendering, the overall structure becomes more regular, with partially improved building shapes. However, local artifacts such as shearing effects and boundary discontinuities still persist in complex regions.

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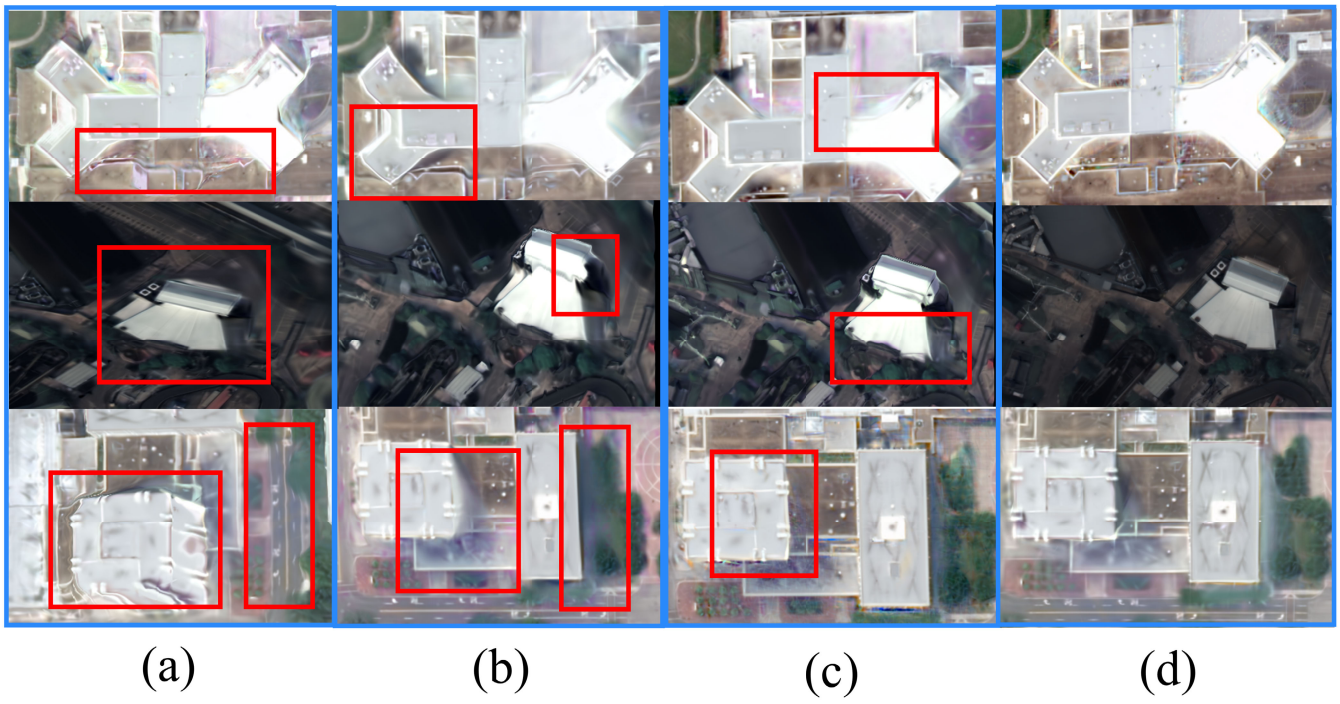


Fig. 12. Qualitative results demonstrating the step-by-step enhancement from the baseline EOGS to the full Tortho–SatGS framework. (a)–(d) Obtained by progressively adding the SfM-based Gaussian initialization, the orthographic rendering module, and the VGG-based perceptual supervision to the baseline. Red boxes highlight key regions, showing how orthographic rendering, SfM initialization, and VGG perceptual loss progressively eliminate geometric distortions and improve visual consistency.

With SfM-based initialization, geometric structures become significantly more stable, with clearer building outlines and improved consistency in occluded areas. Distortions observed in dense urban regions are largely mitigated, which is consistent with the improvement in MAE.

The addition of the VGG perceptual loss further enhances visual quality without altering geometry, leading to sharper textures, smoother boundaries, and improved color consistency, particularly in shadowed and low-texture regions.

Overall, by progressively incorporating the orthographic rendering strategy, the framework is explicitly adapted to the TDOM task, effectively eliminating perspective-induced distortions and improving both geometric and radiometric consistency. The ablation results further indicate that the two modules play complementary roles. The SfM-based initialization primarily enhances geometric accuracy and computational efficiency by providing a compact and reliable structural prior, whereas the VGG perceptual loss mainly improves perceptual fidelity by compensating for the limitations of pixel-wise supervision in weakly textured regions. Their combination achieves a better balance among geometric correctness, rendering efficiency, and visual quality.

2) Effect of SfM-Based Sparse Initialization: To further investigate whether the benefit of the proposed initialization arises from the geometric prior itself or simply from the number of initial primitives, we conduct additional experiments by supplementing the SfM-based sparse point cloud with randomly sampled points of varying densities. Specifically, within the axis-aligned bounding box of the original SfM sparse point cloud, we uniformly sample 4×10^4, 1.2×10^5, and 2×10^5 random points, respectively, and assign each point the mean color of the original sparse point cloud. The resulting models are evaluated in terms of runtime, MAE, PSNR, SSIM, and LPIPS.

The results (Table VIII) show that the proposed SfM-based initialization has a clear advantage in geometric reconstruction. As the number of randomly initialized points increases, PSNR and SSIM generally improve, while LPIPS decreases, indicating enhanced appearance fitting capability. However, MAE exhibits a monotonic increase in most scenes. This suggests that simply increasing the number of primitives can improve image reconstruction quality, but does not lead to better geometric accuracy. Therefore, the effectiveness of the proposed initialization mainly stems from the reliable geometric prior embedded in the SfM sparse structure, rather than from denser point initialization alone. This observation is also important for distinguishing Tortho–SatGS from the original EOGS, which initializes Gaussian primitives through random sampling in the NDC space.

3) Analysis of Perceptual Loss Weight: Unlike the SfM-based initialization, which primarily provides geometric constraints, the VGG perceptual loss mainly acts on appearance modeling and improves perceptual quality by enhancing texture and structural representation. To further evaluate its effect on rendering quality, we systematically compare PSNR, SSIM, and LPIPS under different perceptual loss weights, with $\alpha \in \{0, 0.01, 0.02, \ldots, 0.15\}$.

As shown in Fig. 13, the overall performance exhibits a trend of initial improvement followed by stabilization or slight degradation after introducing the perceptual loss. Within the

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TABLE VIII
ABLATION STUDY ON RANDOM POINT CLOUD INITIALIZATION WITH DIFFERENT DENSITIES

Scene	Random Point Cloud	Runtime (h)	MAE ↓	PSNR ↑	SSIM ↑	LPIPS ↓
JAX_004	0	0.28	1.2857	26.130	0.806	0.419
	40000	0.39	1.8387	30.049	0.880	0.383
	120000	0.46	2.1417	30.953	0.916	0.363
	200000	0.51	2.2660	31.445	0.930	0.338
JAX_068	0	0.29	1.1171	25.221	0.661	0.422
	40000	0.40	1.7695	27.174	0.886	0.288
	120000	0.50	2.1590	27.971	0.905	0.259
	200000	0.55	2.3914	28.376	0.913	0.206
JAX_214	0	0.31	1.7113	21.695	0.832	0.452
	40000	0.45	3.0290	25.663	0.850	0.230
	120000	0.55	4.8736	26.248	0.868	0.210
	200000	0.59	5.7186	26.484	0.877	0.198
JAX_260	0	0.25	1.5822	24.138	0.764	0.464
	40000	0.35	3.4713	26.593	0.806	0.316
	120000	0.44	3.9598	27.236	0.848	0.268
	200000	0.50	4.2647	27.798	0.865	0.245

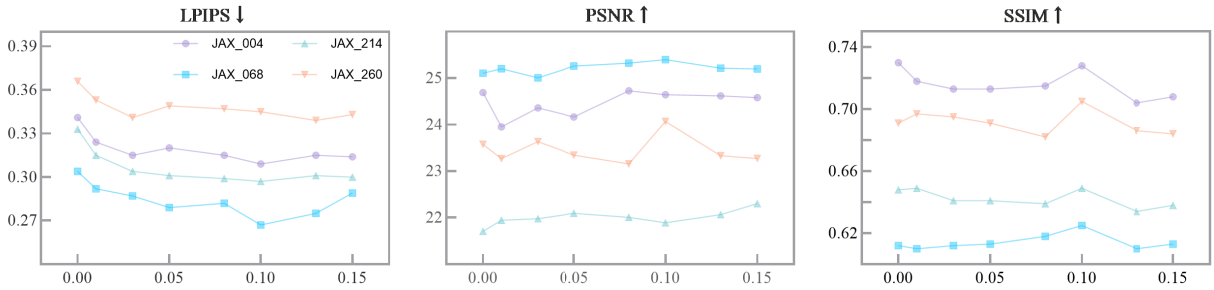


Fig. 13. Model quality under different perceptual loss weights. We compare results obtained without VGG perceptual loss and with perceptual loss weights of 0.01, 0.03, 0.05, 0.08, 0.10, 0.13, and 0.15 in terms of SSIM, PSNR, and LPIPS.

range of $\alpha \in [0.03, 0.10]$, PSNR and SSIM improve, while LPIPS decreases markedly, indicating that the three metrics jointly reach favorable levels across different scenes. When $\alpha > 0.10$, PSNR and SSIM begin to decline slightly in some scenes, and LPIPS no longer decreases further and may even rise marginally. This behavior suggests that a moderate perceptual loss weight is beneficial for improving structural consistency, texture representation, and perceptual similarity, whereas an excessively large weight drives the model to overemphasize high-frequency textures, thereby undermining photometric consistency and increasing pixel-level reconstruction errors.

Considering the combined behavior of all three metrics, we identify the optimal weight range as $\alpha = 0.05$–0.10. Within this interval, the model effectively enhances texture detail and structural representation while preserving photometric consistency, thereby achieving a favorable balance between visual quality and quantitative performance.

V. DISCUSSION

A. Geometric Correctness Under Orthographic Imaging Constraints

Experimental results indicate that Tortho–SatGS provides improved geometric fidelity in true digital orthophoto generation, with particularly clear advantages in building-dense urban scenes. This improvement primarily stems from the explicit enforcement of orthographic imaging constraints within the rendering stage. Unlike conventional orthorectification pipelines, which rely on external DSM/DEM data and visibility analysis that are highly sensitive to elevation errors and occlusion misclassification, the proposed framework renders orthophotos through an exact orthographic transformation of an explicit 3-D scene representation. By replacing perspective projection with parallel ray casting during Gaussian rasterization, Tortho–SatGS effectively eliminates scale variation and facade leakage induced by tilted satellite viewing geometry. As a result, structural artifacts such as roof stretching, boundary bending, and residual side facades are substantially suppressed. These improvements are consistently observed across different land-cover types and acquisition geometries, indicating that the proposed orthographic rasterization strategy provides a reliable mechanism for enforcing true orthophoto geometric consistency without reliance on auxiliary terrain or building models. It is worth noting that NeRF-based methods achieve higher SSIM and lower LPIPS values on certain scenes in the DFC2019 dataset, reflecting stronger reconstruction fidelity under relatively well-conditioned imagery. In contrast, the 3DGS-based reconstruction adopted in this work demonstrates more stable geometric behavior across heterogeneous scenes.

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B. Radiometric Consistency and the Role of Perceptual Supervision

Radiometric inconsistency remains a persistent challenge in satellite orthophoto generation, particularly in shadowed and low-texture regions caused by limited solar angles and complex illumination conditions. The experimental results show that incorporating a VGG-based perceptual loss effectively improves texture coherence and visual completeness in such weakly constrained areas. Unlike pixel-wise losses, which primarily enforce local photometric similarity, the perceptual loss operates in a high-level feature space and introduces a global appearance prior that complements pixel-level supervision. This design encourages semantically consistent texture reconstruction in regions where multiview observations provide insufficient radiometric constraints. Importantly, the perceptual loss is applied only at the image rendering level and does not directly modify the underlying 3-D Gaussian geometry, ensuring that visual improvements are achieved without compromising geometric correctness.

The ablation results further indicate that the perceptual loss weight must be carefully balanced. Moderate weights lead to improved visual quality and perceptual similarity, while excessively large weights may overemphasize high-frequency textures and introduce local photometric inconsistencies. These observations suggest that perceptual supervision serves as an effective auxiliary constraint for enhancing orthophoto visual quality when used in conjunction with geometric and pixel-level objectives.

C. Efficiency and Scalability for Large-Scale Satellite Scenes

A key advantage of Tortho–SatGS lies in its favorable tradeoff between reconstruction accuracy and computational efficiency. By avoiding ray-based volumetric integration and leveraging GPU-accelerated rasterization, the proposed method achieves substantial acceleration during both training and inference compared with NeRF-based approaches. The explicit geometric modeling paradigm of 3DGS uses a fast differentiable rasterizer to generate high-resolution orthophotos while controlling memory usage. This efficiency is particularly beneficial for large-scale satellite mapping scenarios, where computational resources and processing time are critical constraints. As demonstrated in the efficiency analysis, Tortho–SatGS enables practical and scalable true orthophoto generation on commodity GPU hardware.

D. Limitations

Despite its demonstrated robustness, Tortho–SatGS has certain limitations. This is primarily due to the underlying assumption of the RPC model, which approximates a perspective projection within a limited spatial extent. When the spatial coverage becomes too large, this approximation may no longer hold, leading to geometric inconsistencies. Therefore, directly increasing image resolution for large-area scenes is not a straightforward extension. To address this limitation, a potential solution is to adopt a divide-and-conquer strategy, where large-scale satellite imagery is partitioned into smaller regions for independent orthophoto generation, followed by seamless mosaicking. This direction will be explored in future work. In addition, minor brightness inconsistencies may still appear near object boundaries under complex illumination transitions, primarily due to residual coupling between reflectance properties and illumination effects.

VI. Conclusion

This article presents Tortho–SatGS, a novel TDOM generation framework designed for multiview satellite imagery. By integrating 3DGS with satellite-adaptive geometric modeling and orthographic rasterization, the proposed method enables high-quality TDOM generation without reliance on auxiliary elevation data such as DEM, DSM, or DBM. The explicit point-based representation and orthographic rendering strategy effectively suppress perspective-induced geometric distortions, yielding structurally consistent orthophotos across complex urban and natural scenes. To address persistent radiometric challenges in satellite imagery, a VGG-based perceptual loss is introduced to enhance texture coherence and visual realism, particularly in shadowed and low-texture regions. Extensive experiments on public satellite datasets demonstrate that Tortho–SatGS consistently outperforms conventional high-based and NeRF-based approaches in geometric accuracy, radiometric consistency, and fine-detail preservation. Moreover, benefiting from the rasterization-based efficiency of 3DGS, the proposed method achieves substantial acceleration during both training and rendering, enabling orthophoto generation on commodity GPU hardware.

Overall, Tortho–SatGS provides an efficient and reliable pure visual solution for satellite TDOM generation, advancing the state of the art in satellite orthorectification and large-scale geospatial data production.

Disclosures

The authors declare no conflicts of interest related to this research.

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